

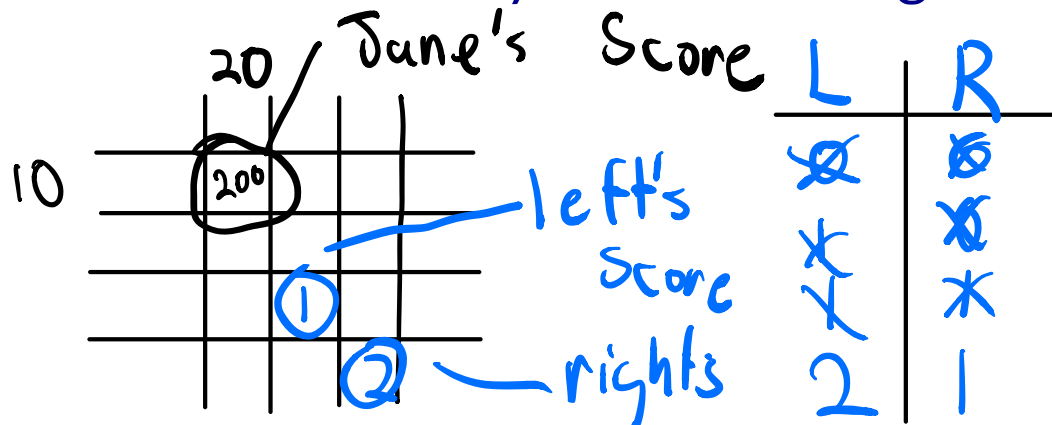
Chapter 4 - Variables

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01/26/2021

Variables

- A metaphorical bucket
- Computers have memory, allowing you to store information for later use
- *Variables* are pointers to a location in computer's memory
- Return of Graph Paper
- Variable values can be altered or *vary*
- What information do you need for a game of pong?



Variables

- Variables hold *primitive* values or *references* to *objects* and *arrays*.
- *Primitive* variables usually refer to holding a singular piece of information, like:
 - Whole numbers (1, 2, 3, 4)
 - Referred to as an *integer*, with *int* as the keyword.
 - Decimal Numbers (3.45, 2.5)
 - *Floating-Point Values*, with *float* as the keyword.
 - Characters (Single letters like “a”, “b”, “c”)
 - Stored with *char* and using single quotes (')

Primitive Types

- *boolean*: true or false
- *char*: a character
- *byte*: a small number, -128 to 127
- *short*: larger number, -32,768 to 32,767
- *int*: a big number, -2,147,483,648 to 2,147,483,647
- *long*: huge numbers
- *float*: a decimal number
- *double*: decimal numbers with more decimal places

Variable Assignment

- Once a variable is created, it can be assigned a value.
 - Known as *initializing*

```
int count; count = 50;
```

Type Name Assignment

```
int count = 50;
```

```
int num = 0;
```

Naming Variables

- Avoid using words that appear elsewhere in Processing (like *setup* or *mouseX*)
- Use names that describe what the variable is storing.
 - If you're going to keep track of a player's score, name the variable `score`, not `cat`.
- Start variables with a lowercase letter and join words with capitals. `janesScore`.
 - This will become second nature with some experience.
 - This is known as *camelCase* if you're curious

Variable Assignment

- Once a variable is created, it can be assigned a value.
 - Known as *initializing*

- `int count;`

- `count = 50;`

- One-Line:

- `int count = 50;`

Data Variable

Type Name

Creating Variables

- Variables need a *type* and a *name*.
- Rules for naming:
 - One Word (no spaces)
 - Must start with a letter
 - Can include numbers, just not at the start
 - No punctuation, except the underscore _

~~1num~~ num1 jane_score
~~2num~~ num2

1+1

Naming Variables

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will become

Using Variables

- *mouseX* and *mouseY* are variables!
 - Built-in into Processing
- **Example 4-2**

Questions

- Things to think about before writing a sketch
 - *What data I need to remember for the sketch?*
 - *How do I use that data to draw shapes on the screen?*
 - *How do I alter that data to make my sketch interactive and animated?*

```
int left_score = 0;  
int right_score = 0;
```

Many Variables

- Using variables for every possible thing?
- **Example 4-4**
- *mouseX / mouseY*
 - What are they?

System Variables

- Commonly used *built-in* variables
- **width**
- **height**
- **frameCount**
- **framerate**
- **displayWidth**
- **displayHeight**

System Variables (more)

- **key**
- **keyCode**
- **keyPressed**
- **mousePressed**
- **mouseButton**
- *Avoid overwriting these variables.*
 - You can replace them and will no longer work as expected.
- **Example 4-5**

Random

- *Incremental Development*
- `random()` ;
- **Example 4-6**
- Random returns float, fill expects integers?
 - *casting*
- **Example 4-7**

`int r = 0;`

4. ~~8~~ → 4

`r = random(255);`

`r = int(random(255));`

Variable Zoog

- Example 4-8

Translation

- **Example 3-7**
- So far, all shapes have been drawn relative to the origin point
 - $(0, 0)$ - Top Left of the Display
- `translate()` ;
 - You can change the offset of drawing or
 - The relative point*
- *Chapter 14*, more on that next week